



Mobile phone data policy applications

Session 2

FLOWMINDER.ORG

NOMMON

positium

Overview

This session covers the following topics:

1. Overview of MPD Policy Applications
2. Measuring Sustainable Development Goals (SDG)
3. Population Statistics and Mobility Analysis:
 - a. Dynamic Population Mapping
 - b. Migration & Disaster Management
 - c. Public Health & Epidemiology
 - d. Transport
 - e. Tourism
4. Socioeconomic Applications: Poverty Mapping
5. Conclusions and Recommendations

Overview of MPD policy applications

Many application opportunities

Mobile phone data (MPD) is a valuable source for the production of official statistics in a variety of policy domains where it is critical to provide reliable and up-to-date indicators on population's presence and mobility patterns.



**Production of highly detailed,
accurate statistics**



**Fast and cost-effective
collection of data**



**Increase the frequency of
updated information**

The exploration of MPD

Many National Statistical Offices (NSOs) have launched activities aimed at using MPD. To support these efforts, the UN defines 6 areas for the exploitation of MPD:

1) Tourism statistics

2) Migration statistics

3) Census and
dynamic population

4) Displacement in
disaster context

5) Information society

6) Transport and
commuting statistics

<https://unstats.un.org/bigdata/task-teams/mobile-phone/index.cshtml>

 **UN Big Data**

The exploration of MNDs

For this course, three main groups of applications are considered:

1. Measuring Sustainable Development Goals (SDG)
2. Population Statistics and Mobility Analysis
3. Socioeconomic Applications: Poverty Mapping

Measuring Sustainable Development Goals (SDG)

Indicators and Use Cases

Indicator: proportion of individuals who used internet from any location in the previous three months (SDG 17.8.1)

Use cases:

- Sociodemographic analysis of Internet access

Indicator: proportion of the population covered by a mobile network (SDG 9.c.1)

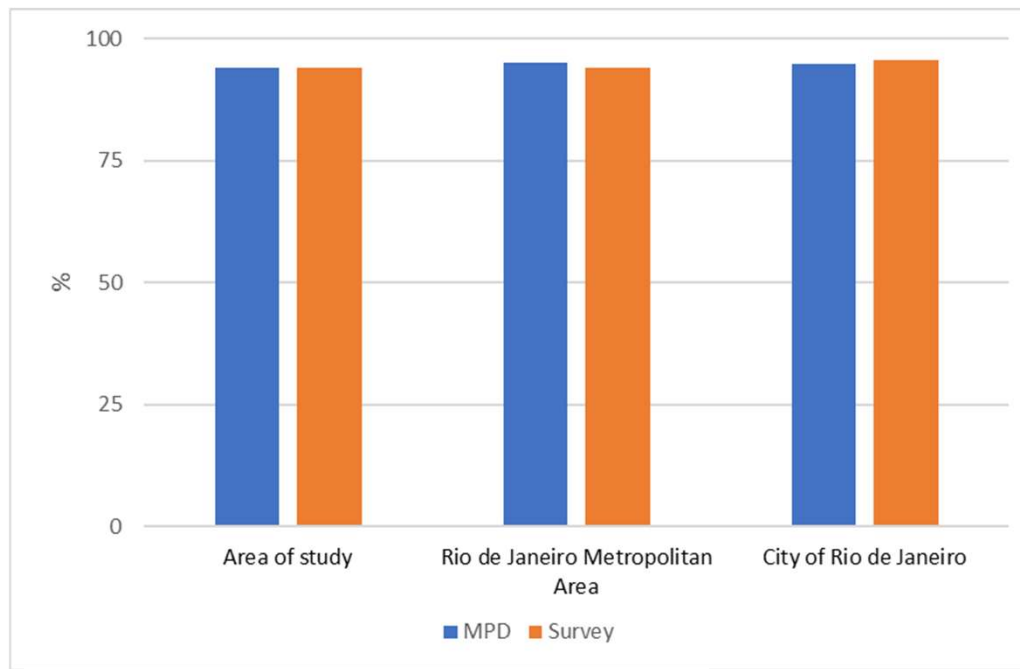
Use cases:

- Infrastructure and resource planning

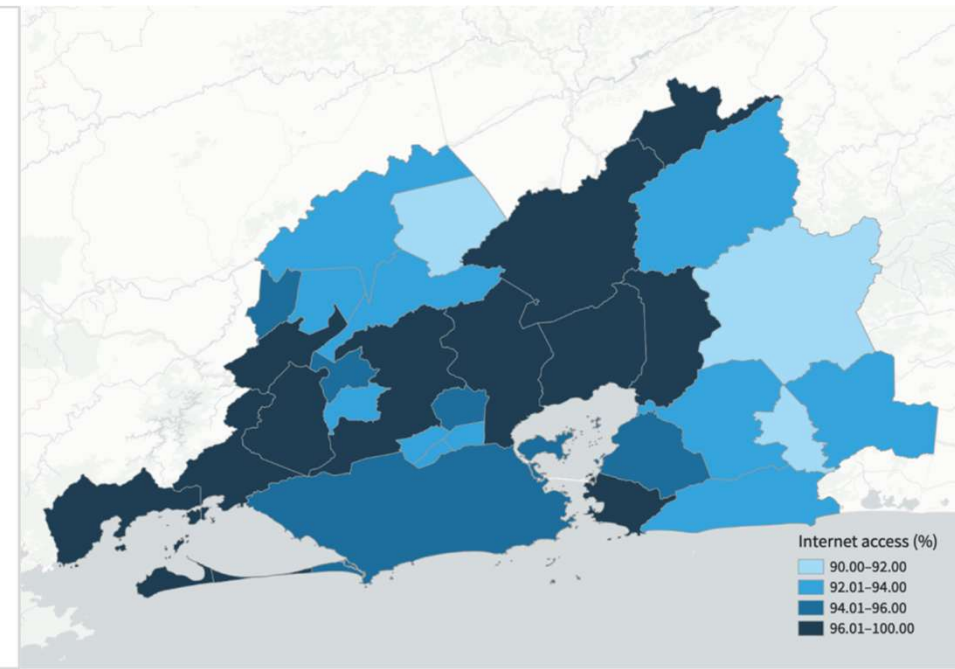
Methodological guide on the use of MPD - information society SDG indicators:
<https://unstats.un.org/wiki/display/MPDMIS>

Proportion of individuals using Internet

Example: Rio de Janeiro, Brazil



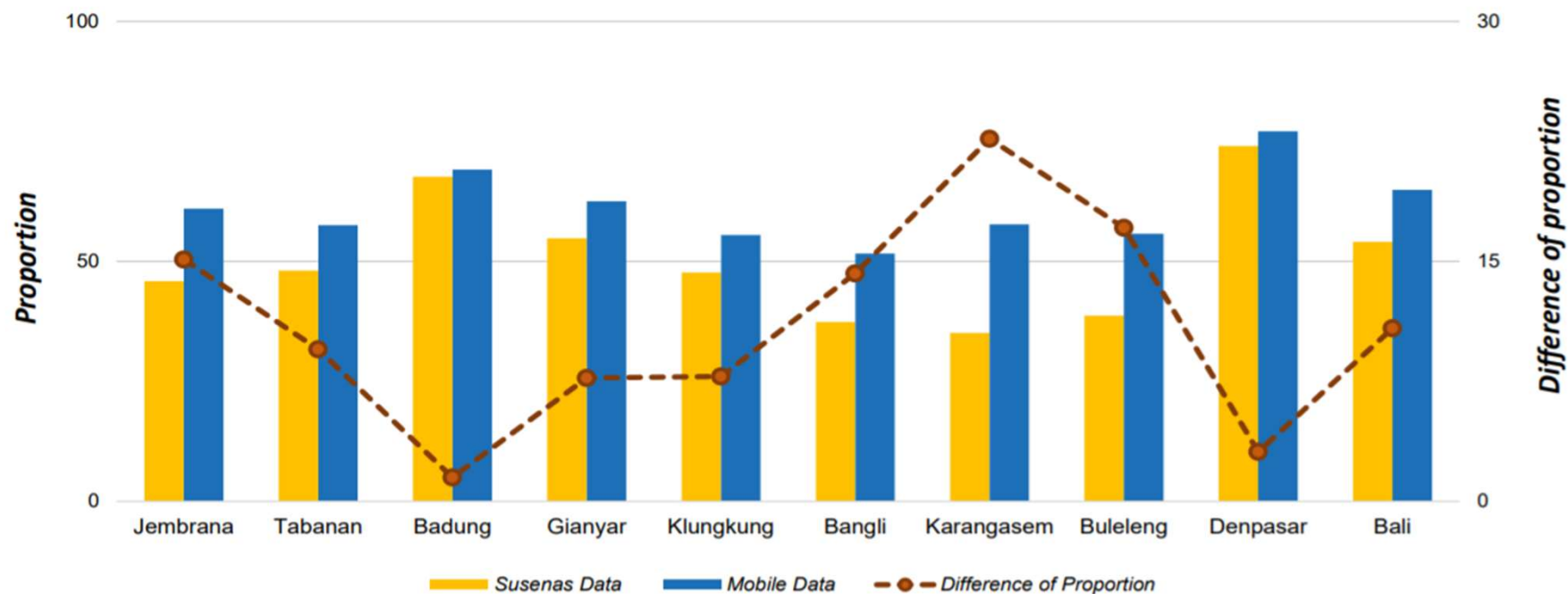
Source: Brazilian Institute of Geography and Statistics (IBGE)



Source: Positium

Proportion of individuals using Internet

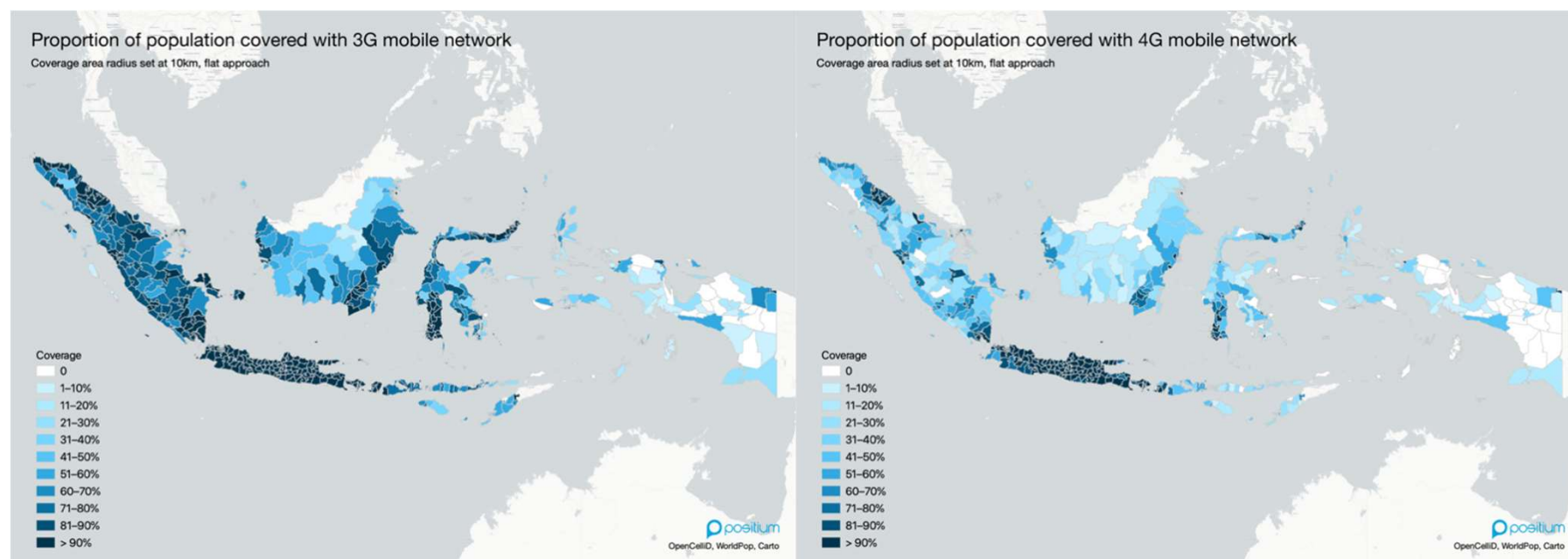
Example: Bali, Indonesia



Source: BPS Indonesia. The proportion differences of internet users (17.8.1) in Bali, Indonesia, between mobile phone data and survey data from the National Socio-Economic Household Survey (Susenas)

Proportion of population covered by a mobile network

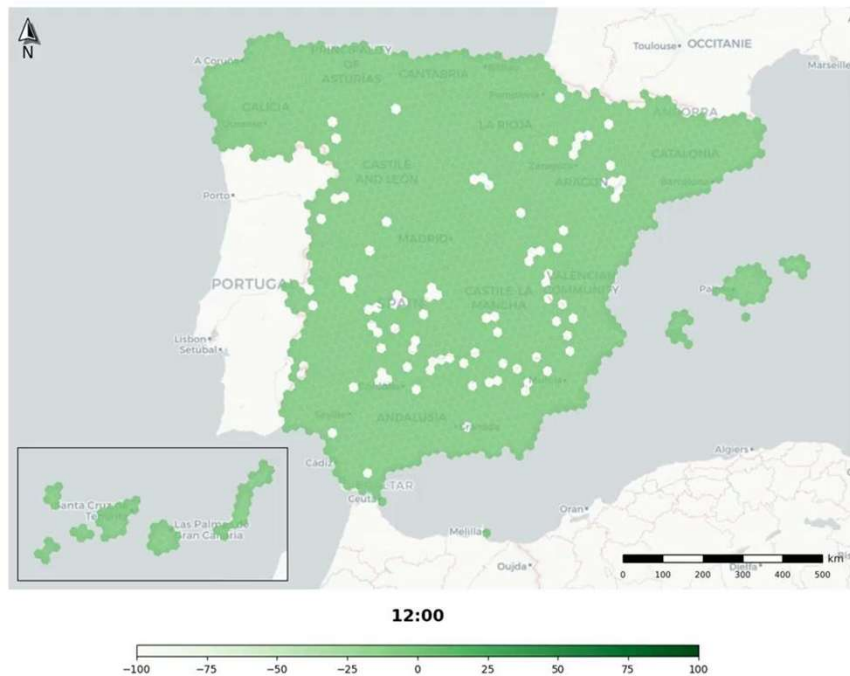
Example: Indonesia



Source: Positium

Access to internet and mobile connection

Example: Electric blackout, Spain



Source: Nommon

- Massive electric blackout in the Iberian region (Spain and Portugal)
- Mobile phone data made it possible to understand the timeline of the blackout and how it affected people's lives compared to a normal day.

Population Statistics and Mobility Analysis

a. Dynamic Population Mapping

Indicators and Use Cases

Indicator: number of people present in each area of the territory at each moment in time

Use cases:

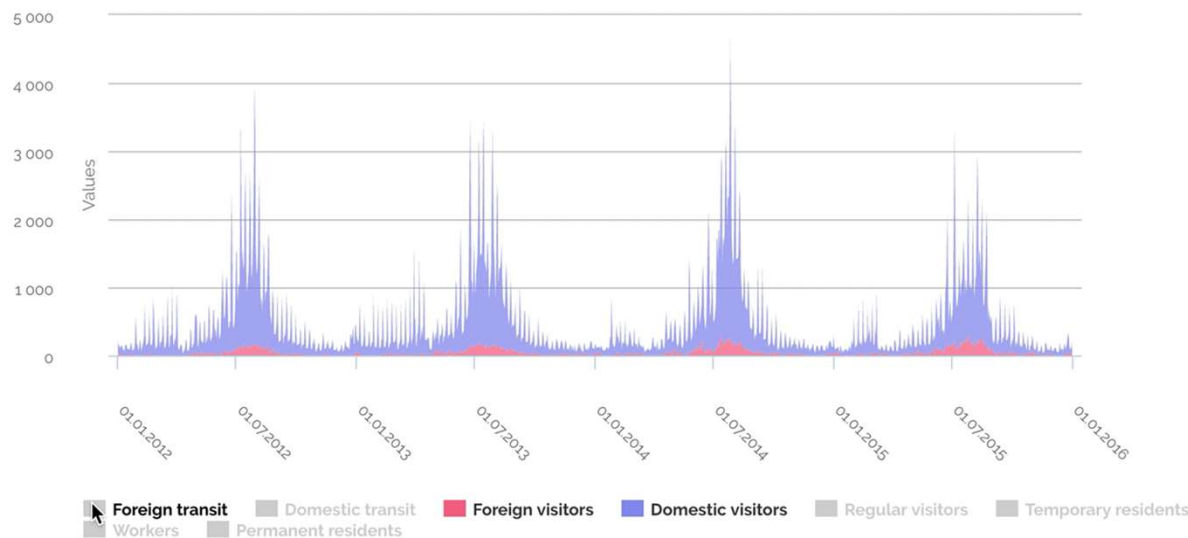
- Resident population mapping
- Daytime and “de facto” population mapping
- Event management
- Infrastructure and resource planning
- Creation of dynamic sample frames for surveys
- Census

Methodological guide on the use of mobile phone data: Dynamic population mapping,
<https://unstats.un.org/wiki/display/MPDDPM>

De facto population

Example: Estonia

Identification of residents, temporary residents, visitors, and other people who stay in an area.



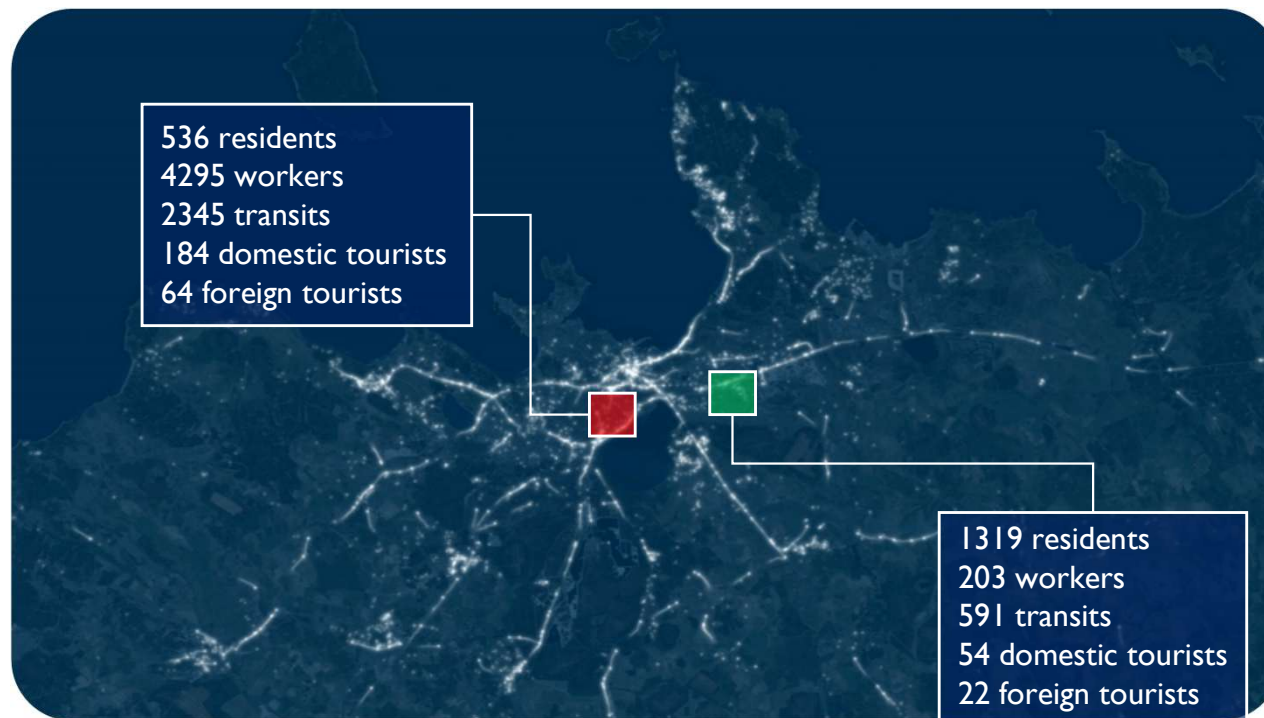
Users

- National and local governments
- Law enforcement
- Disaster management agencies
- ...

Source: Positium

Dynamic Population Mapping

Example: Estonia



Source: Positium

Census

Opportunities to complement the census during:

- Census preparation and execution (e.g., establishing survey frames)
- Intercensal period (e.g., update of population projections)
- Measuring possible undercounted populations (e.g., nomadic populations)
- Adding features (e.g., daytime population, migration or commuting)

And to carry out the Census through an innovative approach:

- Data-driven census combining administrative and big data signs of life even where population registers are non-existent/non-complete.

Example: Using Mobile Positioning for Improving the Quality of Register Data, Estonia:

<https://www.statistikuasociacija.lv/workshop2018/files/papers/BNU2018-S%C3%B5stra.pdf>

Population Statistics and Mobility Analysis

b. Migration & Disaster Management

Indicators and Use Cases

Indicator: residential movements during a period of time

Use cases:

- Migration statistics

Indicator: temporal displacement of people during a period of time

Use cases:

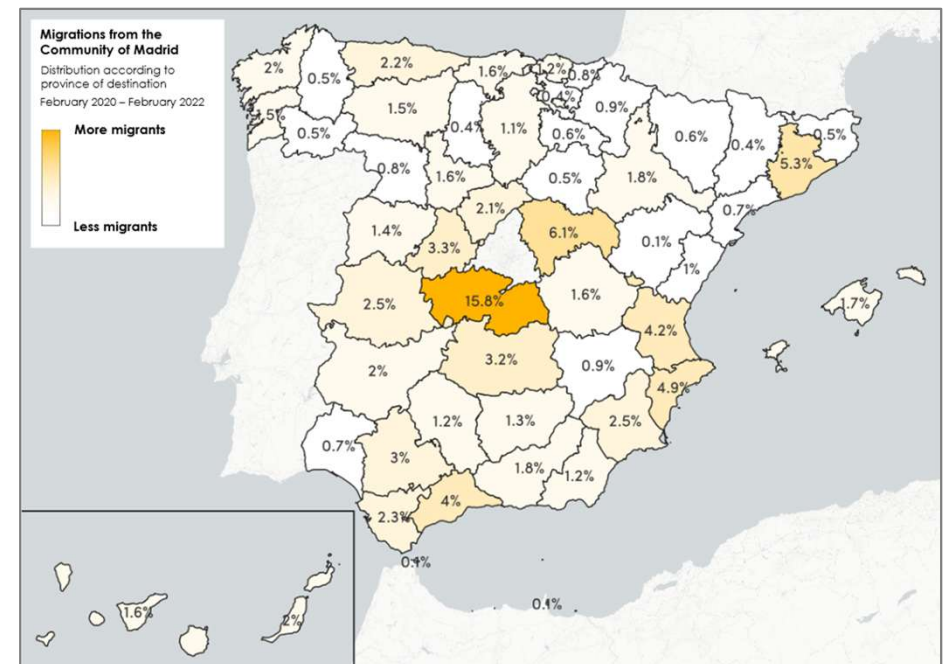
- Disaster and displacement statistics

Methodological guide on the use of mobile phone data: Migration and Displacement and disaster statistics,
<https://unstats.un.org/wiki/display/MPDMS> / <https://unstats.un.org/wiki/display/MPDDS/>

Residential movements

Example: Migration during COVID-19 in Spain

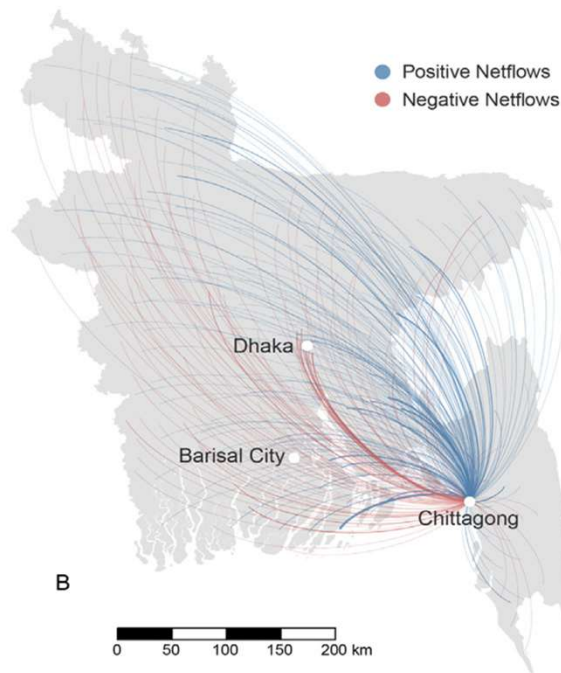
- Increase in migration from the Region of Madrid to almost all regions, with a notable increase in the surrounding provinces
- The population that remained in Madrid dropped by around 10%.
- The main receiving regions were Andalusia, Valencia, and Castile and León.



Source: Nommon

Climate migration

Example: Bangladesh



Source: Lu *et al.* Global Environmental Change. 2016

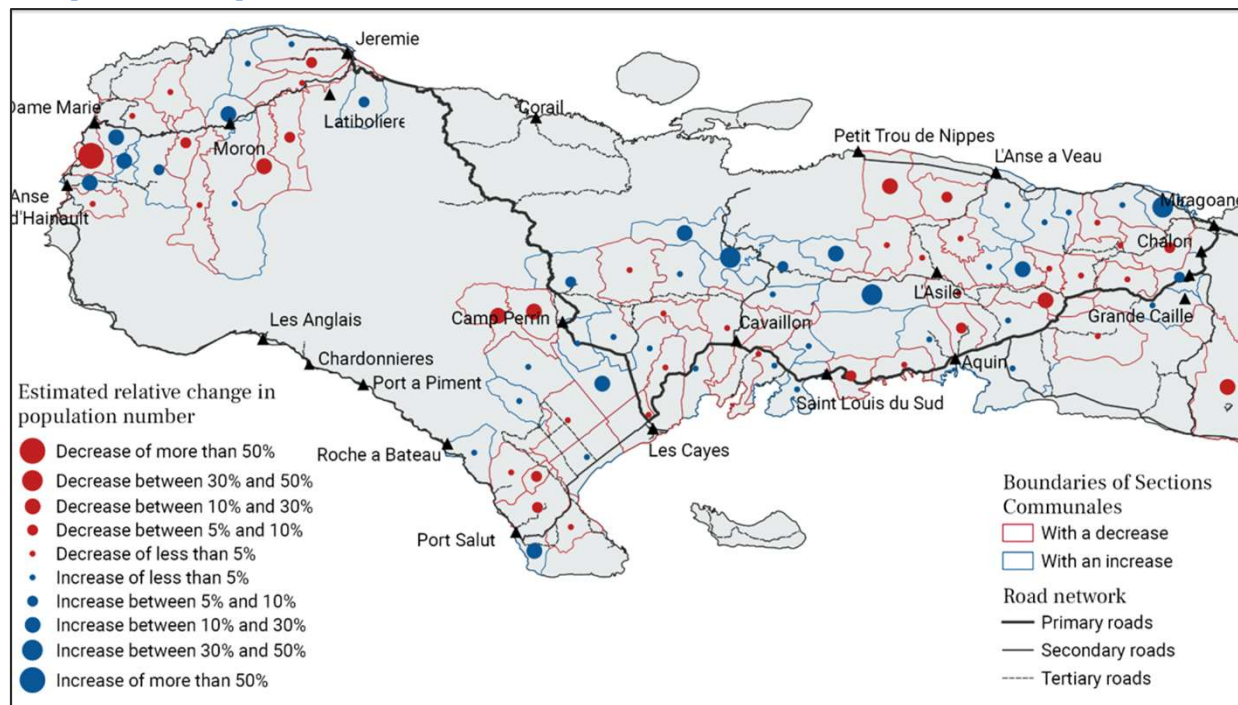
Identifying and quantifying migration caused by climate stress, to design programmes for mitigation and adaptation.

MPD was used to quantify migration episodes from climate stressed areas, at local and national scales

- Incidence
- Direction
- Seasonality
- Duration

Disaster displacement

Example: Earthquake reports in Haiti



Source: Flowminder 2021, Haiti Earthquake Reports

Population Statistics and Mobility Analysis

c. Public Health & Epidemiology

Indicators and Use Cases

Indicator: Population exposure to environmental health factors

Use cases:

- Exposure to pollution
- Exposure to noise
- Exposure to heatwaves

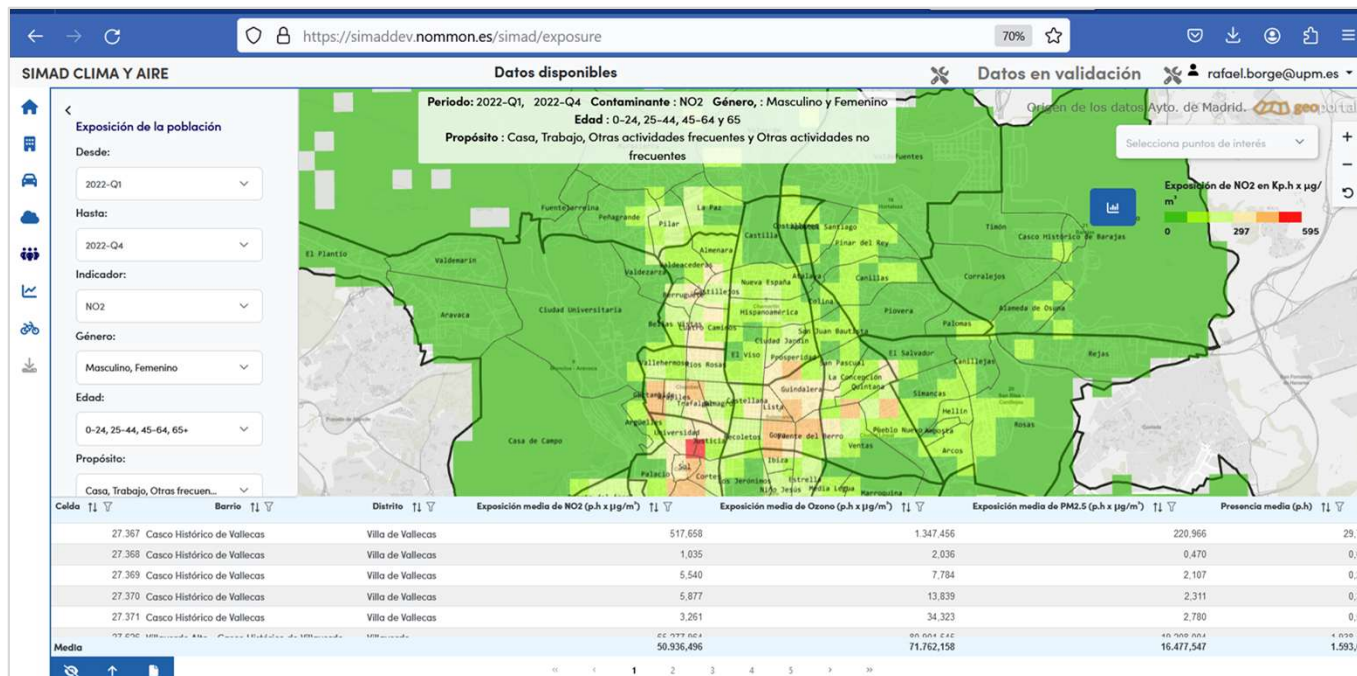
Indicator: Social contact

Use cases:

- Disease spreading / epidemiology

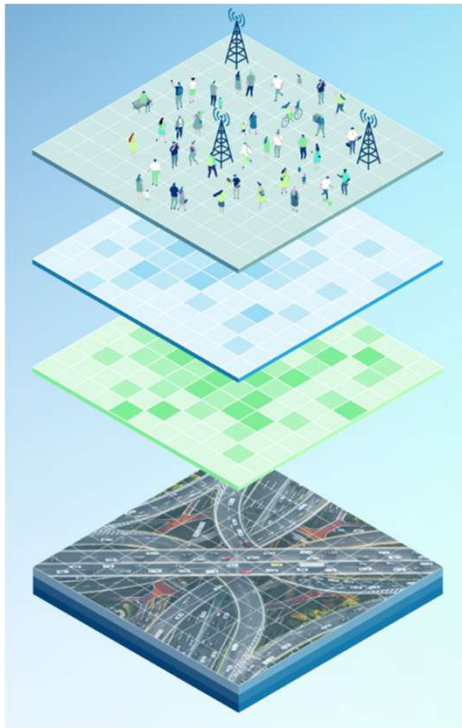
Exposure to pollution

Example: Air Quality in Madrid, Spain



Source: Nommon / Madrid City Council

Disease spreading / epidemiology



Example: MePreCISa Project, Spain

- Open Cloud Platform for Managing Health Crises in Complex Scenarios fusing MPD with different data sources
- Modeling Exposure to Contaminants and Disease Spreading
- Three use cases:
 - Mobility, Air Quality and Health
 - Social Contact and Spread of Epidemics
 - Mobility and Wastewater

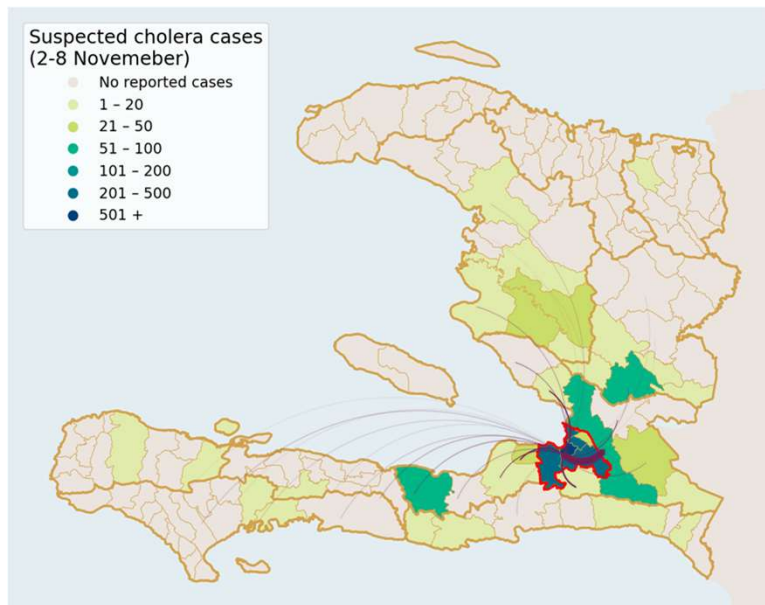
Source: Barcelona Supercomputing Center, <https://meprecisa.com/>

Disease spreading / epidemiology

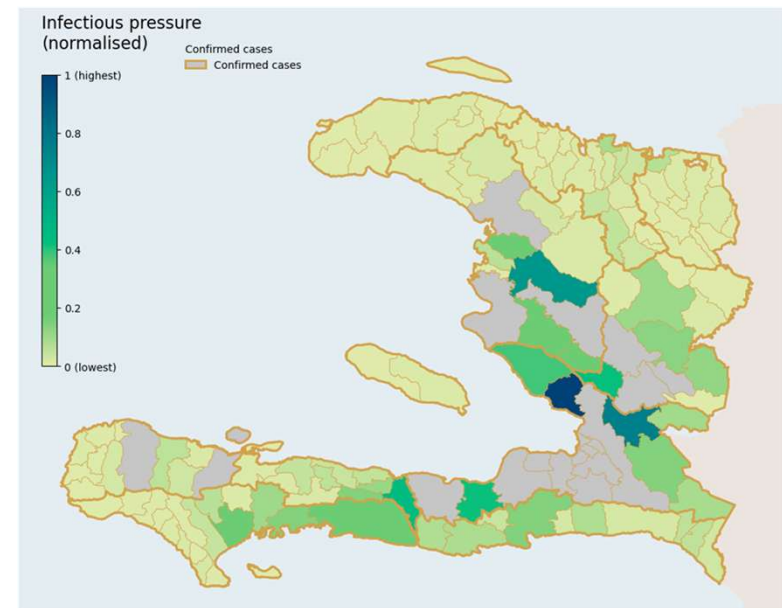
Example: Cholera in Haiti

Identifying where the outbreak is likely to spread to in order to prioritise surveillance and resource allocation

Flows from areas with high incidence in Port-au-Prince → Infectious **pressure** on unaffected areas



Source: Flowminder



Population Statistics and Mobility Analysis

d. Transport

Indicators and Use Cases

Indicator: Origin-Destination matrices by transport mode, trip frequency and trip distribution

Use cases:

- Transport planning
- Transport modeling
- Transport concessions
- Traffic management

Methodological guide on the use of mobile phone data: Transport Statistics,
<https://unstats.un.org/wiki/display/MPDT>

Understanding travel behaviour: many use cases



Mobility planning



Tourism management



Transport concessions



Shared mobility



Traffic management



Competition studies



Transport hubs

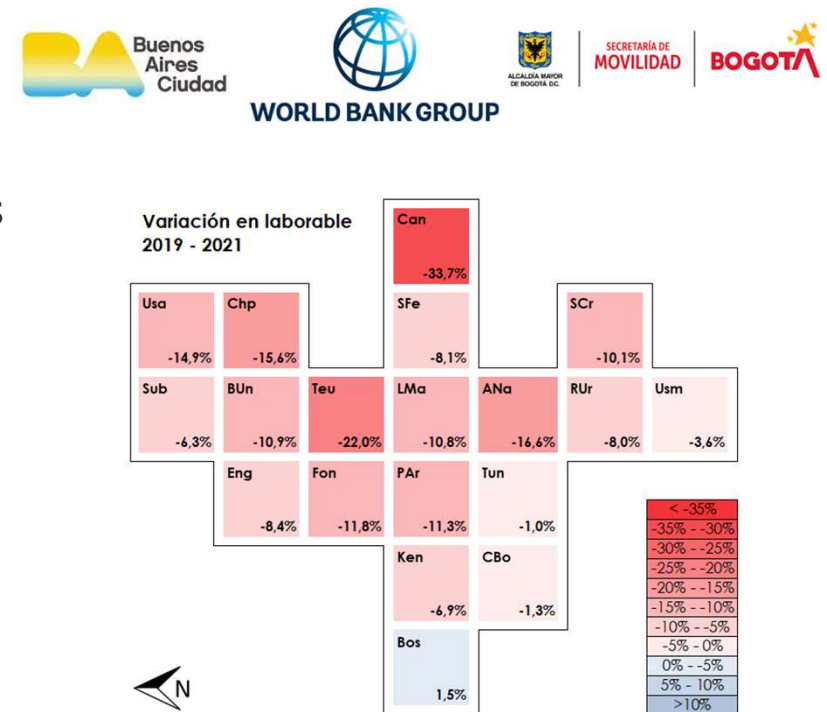


Mobility monitoring

Monitoring travel demand

Example: Latin America cities

- Generation of origin-destination matrices for 2019, 2020, and 2021 to evaluate the pandemic's effect on mobility habits in the metropolitan areas of Bogotá (Colombia) and Buenos Aires (Argentina).
- Segmentation by transport mode by fusing MPD with survey data and Automated Fare Collection data from public transport authorities.
- Replication of the methodology in Medellín (Colombia), Asunción (Paraguay), Rio de Janeiro, Florianópolis, and Belo Horizonte (Brazil)



Percentage variation of trip generation in Bogotá, Colombia, between 2019 and 2021, for an average working day. Source: Nommon

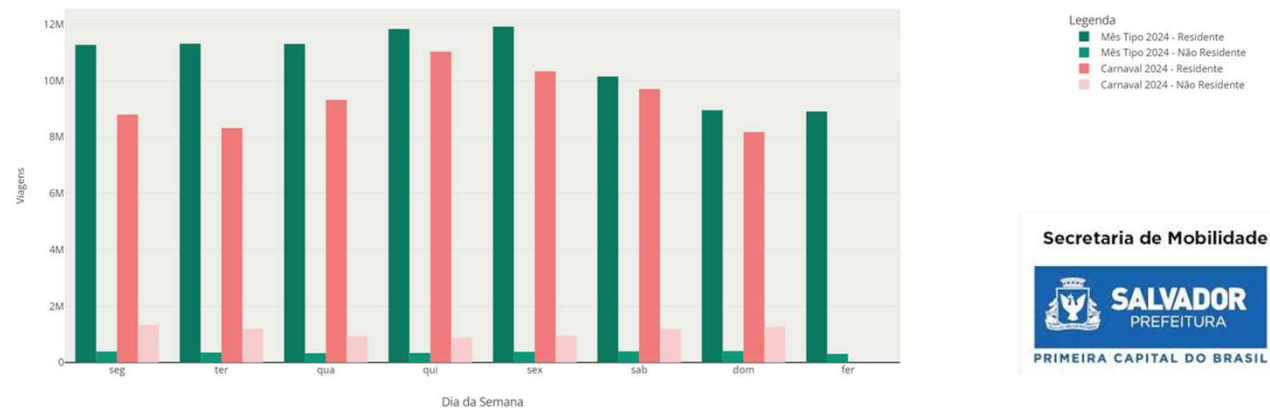
Monitoring travel demand

Example: OD matrices for Salvador, Brazil

Analysis of travel demand in for the local government

- Days of study: 15 days of 2024, covering weekdays, weekends and the Carnival period
- Zoning system: 385 zones
- Segmentation by time of the day, trip purpose, trip distance, sociodemographic profile

Produção total de viagens na RM de Salvador (residentes e não-residentes)

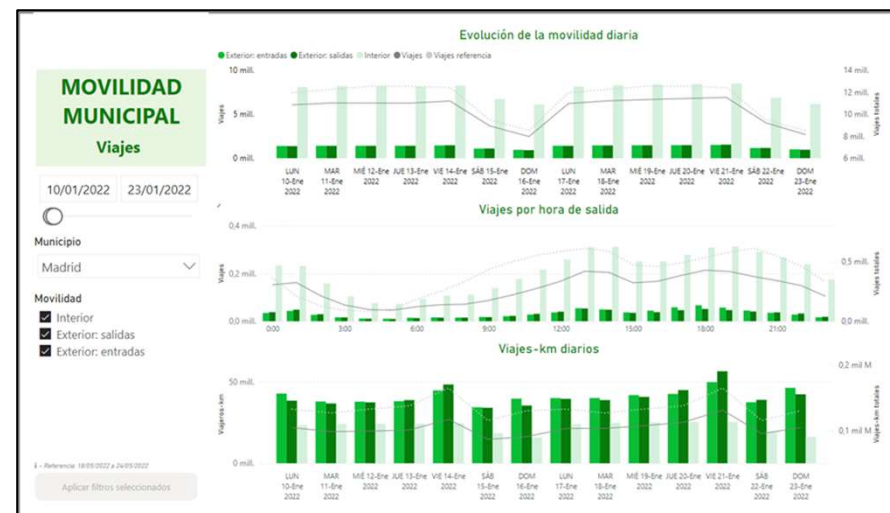


Total number of trips in the city of Salvador, Brazil, for residents and non-residents in an average working day and during Carnival, 2024. Source: Nommon & TTC

Monitoring travel demand

Example: Analysis of passenger mobility in Spain

- Complete analysis of mobility in Spain with over 3,000 study areas:
 - Daily indicators of total mobility flows
 - Monthly indicators by mode of transport
 - Biannual information on road route choices
- Combination with transport supply and demand data for all modes.
- Drastic time and cost reduction, along with an expanded scope, compared to the previous national travel survey.



<https://www.transportes.gob.es/ministerio/proyectos-singulares/estudio-de-movilidad-con-big-data>. Source: MITMS / Nommon,

Population Statistics and Mobility Analysis e. Tourism

Indicators and Use Cases

Indicators: number of tourists (local and foreign) that visit each area during a period of time, number and location of overnight stays, duration of a stay in a specific area or tourist spot

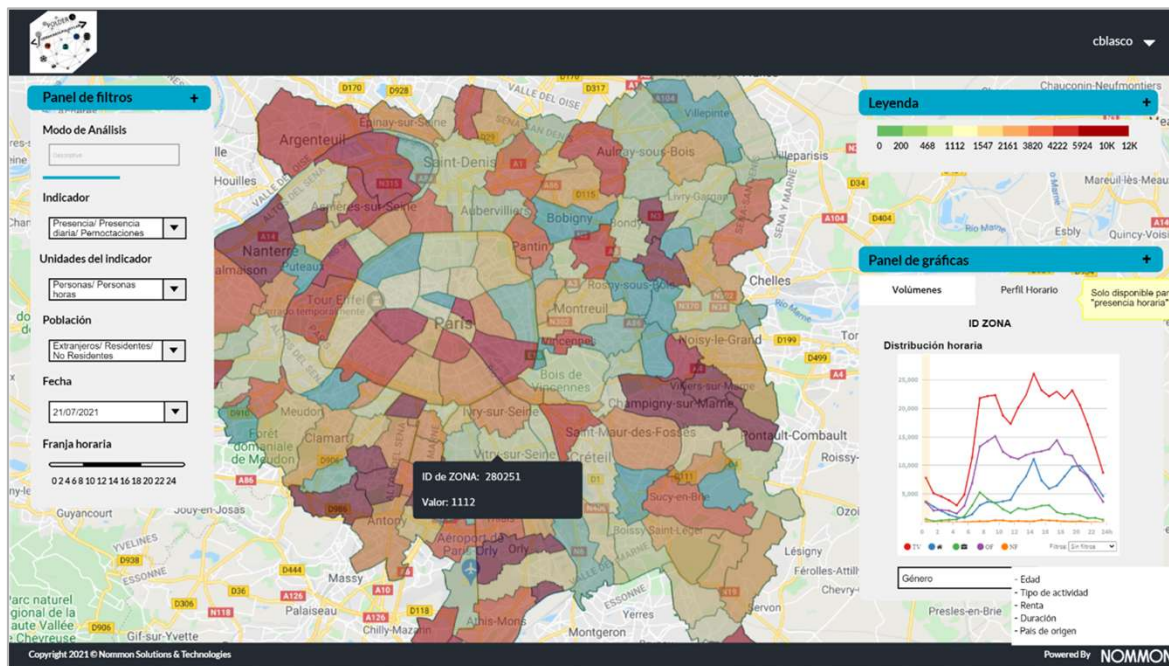
Use cases:

- Tourism statistics
- Crowd management (open and closed spaces)
- Protection of natural and cultural heritage

Methodological guide on the use of mobile phone data: Tourism Statistics,
<https://unstats.un.org/wiki/spaces/MPDTS/overview>

Tourism indicators

Example: Analysis of tourist demand in Madrid, Spain

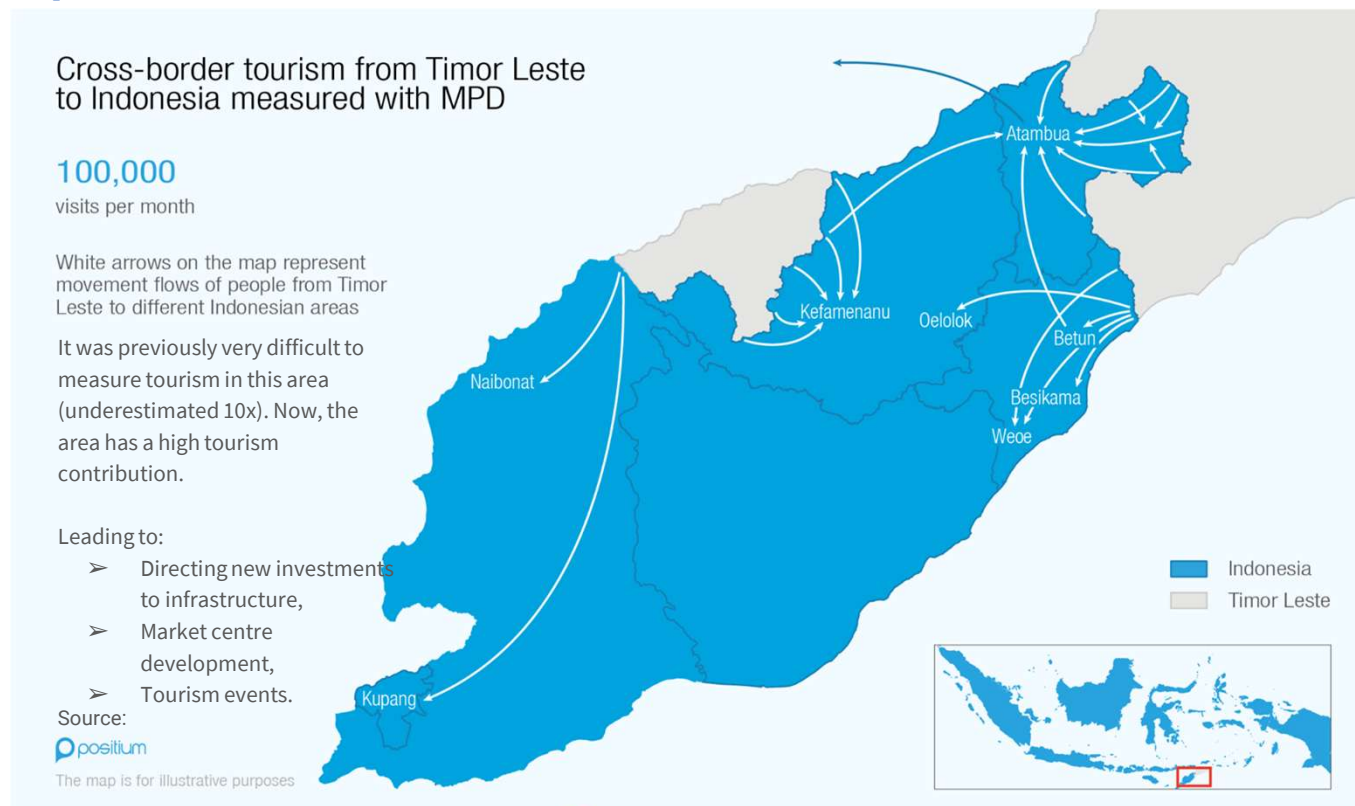


Source: Nommon

- Platform for the analysis of tourism indicators
- Classification of areas according to their tourist activities
- Forecast of the number of overnight stays by foreign tourists

Tourism indicators

Example: Cross-border tourism in Timor Leste



Socioeconomic Applications: Poverty Mapping

Indicators and Use Cases

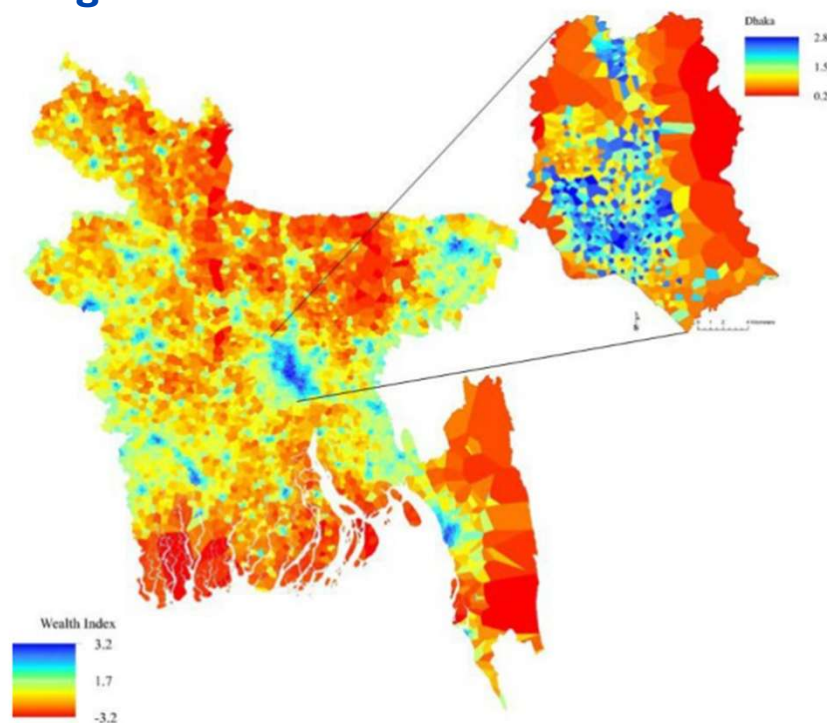
Indicator: Spatial distribution of poverty (fusion of MPD with other data sources)

Use cases:

- Mean wealth index
- Accessibility to basic services

Poverty Mapping

Example: Bangladesh



National level prediction map in Bangladesh for the mean wealth index generated using mobile phone features, remote sensing data, and Bayesian geostatistical models. Red indicates poorer areas, zoom area is the capital city, Dhaka.

National level prediction map in Bangladesh for the mean wealth index generated using mobile phone features, remote sensing data, and Bayesian geostatistical models. Red indicates poorer areas, zoom area is the capital city, Dhaka. Source: Steele et al., Royal Soc. Int. (2017)

Conclusions and Recommendations

Assessing the impact of a given application

- ✓ Does it **address an existing gap** in official statistics?
- ✓ To what extent does it **reduce data collection costs** for a given indicator without compromising quality?
- ✓ To what extent does it **improve the quality** of a given indicator without increasing data collection costs?
- ✓ How effectively can the output indicators support governments in **making more efficient and informed decisions**?



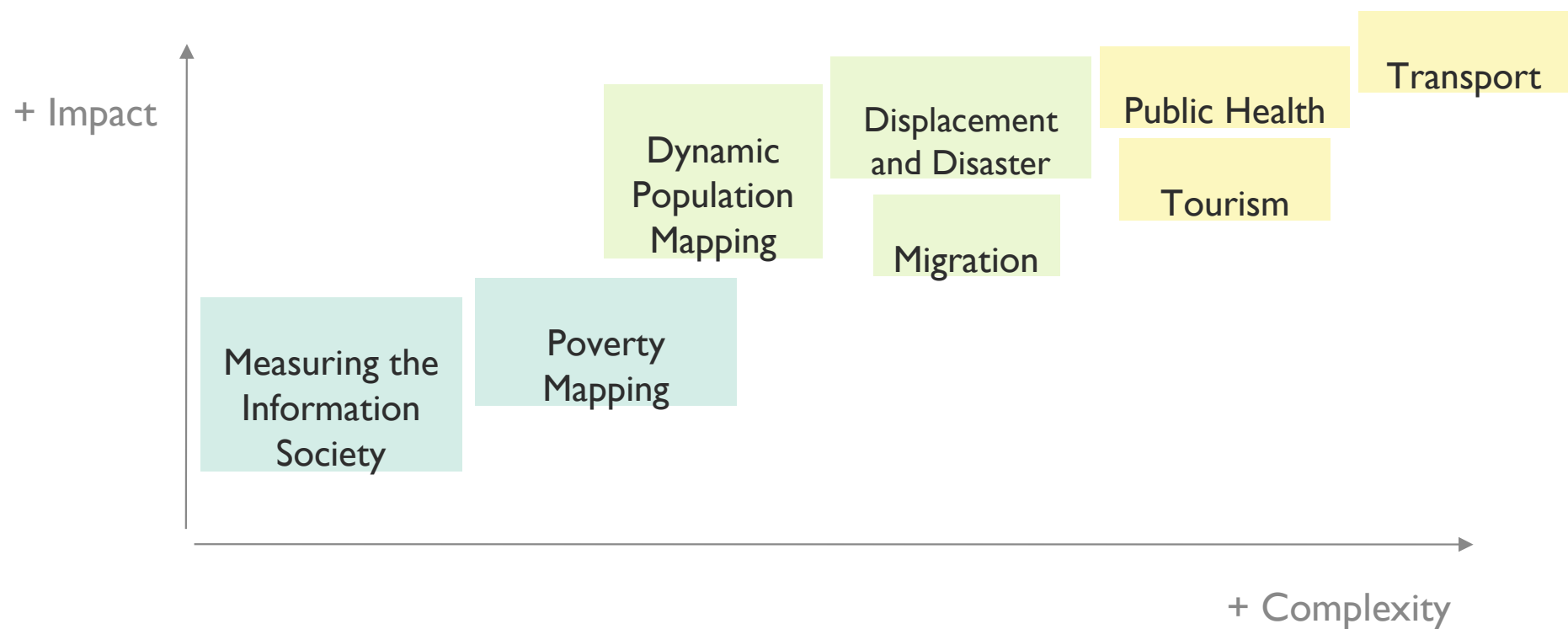
Assessing the complexity of a given application

- ✓ What **characteristics** must the input **MPD** have?
- ✓ How complex are the required **MPD processing algorithms**?
- ✓ Does it need to be **combined with additional data sources**?
- ✓ What **level of accuracy is required** for the output indicators to be useful for decision-making?



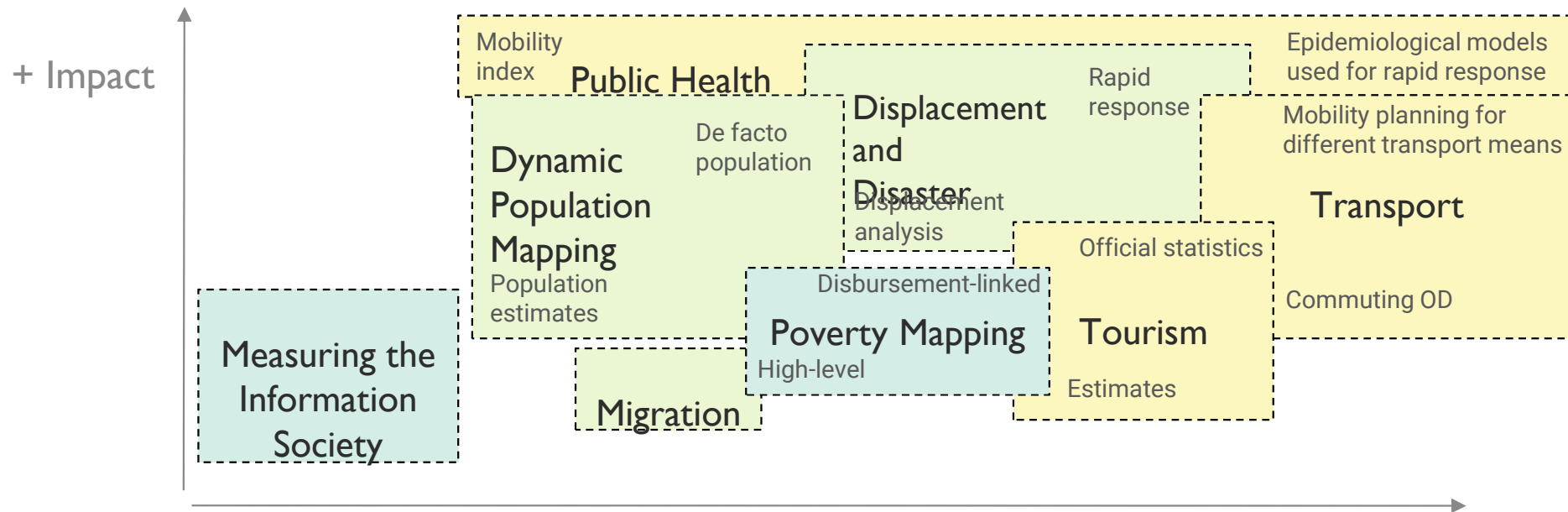
Complexity vs. impact

Mapping the different types of indicators



Complexity vs. impact

Mapping the different types of indicators according to the ranges of complexity and impact various implementations of MPD can have. Examples provided in grey.



Note: The visual is meant to provoke discussion, not to be taken as a definitive guide. The actual impact and complexity depends on the depth of data analysis and use.

Thank you!



The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).



FLOWMINDER.ORG

NOMMON  Official Use Only



Mobile Phone Data for Policy Programme (WB-GDF)
© 2025 by Flowminder, Positium, Nommon, ITU and
World Bank is licensed under CC BY 4.0.

To view a copy of this license, visit
<https://creativecommons.org/licenses/by/4.0/>